Practical 5

1. Loading data

The aim of this practical was to explore a dataset and apply two privacy techniques that we learned about in the module: generalisation and suppression. These methods help reduce the amount of personal detail in a dataset while still keeping the information useful. I used Python to load the German Credit dataset, inspect the attributes, apply changes to the data, and reflect on what the dataset can still tell us after anonymisation.

I began by loading the credit.csv file into a Jupyter Notebook and printing the first few rows to check that the file had loaded properly.I then looked at the size of the dataset, the types of attributes, and the number of unique values. This helped me understand what kinds of information were included and how detailed some of the attributes were.

```
[1]: import pandas as pd

     gc = pd.read_csv("Credit.csv")
     gc.head()
```

[1]:

	Unnamed: 0	Default	checkingstatus1	duration	history	purpose	amount	savings	employ	installment	sex	residence	age	housing	cards	liable	tele	forei
0	1	0	<DM0	6	terrible	goods/repair	1169	DM0-100	7+years	4	male	4	67	own	2	1	yes	forei
1	2	1	DM0-200	48	poor	goods/repair	5951	DM0-100	1-7years	2	female	2	22	own	1	1	no	forei
2	3	0	NoAccount	12	terrible	edu	2096	DM0-100	1-7years	2	male	3	49	own	1	2	no	forei
3	4	0	<DM0	42	poor	goods/repair	7882	DM0-100	1-7years	2	male	4	45	forfree	1	2	no	forei
4	5	1	<DM0	24	poor	newcar	4870	DM0-100	1-7years	3	male	4	53	forfree	2	2	no	forei

[ ]:

2.Purpose generalisation

The purpose attribute originally had several specific categories. To reduce detail, I grouped these values into two broader categories: car and other (non car).Generalising the attribute makes it harder to identify individuals using specific values, but the data is still useful because we can still compare the main types of loan purposes.

```
[5]: gc["purpose"].value_counts()
```

```
[5]: purpose
     goods/repair    495
     newcar          234
     biz             109
     usedcar         103
     edu              59
     Name: count, dtype: int64
```

```
[6]: gc.purpose = gc.purpose.replace(['newcar', 'usedcar'], 'car')
     gc.purpose = gc.purpose.replace(['biz', 'edu', 'goods/repair'], 'other (non-car)')
     gc["purpose"].value_counts()
```

```
[6]: purpose
     other (non-car)    663
     car                337
     Name: count, dtype: int64
```

3. Summary table

I then looked at the summary statistics of the numerical attributes and plotted histograms. This gave me an overview of how the values were spread across the dataset.Even before anonymisation, the dataset already showed patterns, such as the range of loan amounts and the spread of ages.

```
[7]: gc.describe().T
```

[7]:

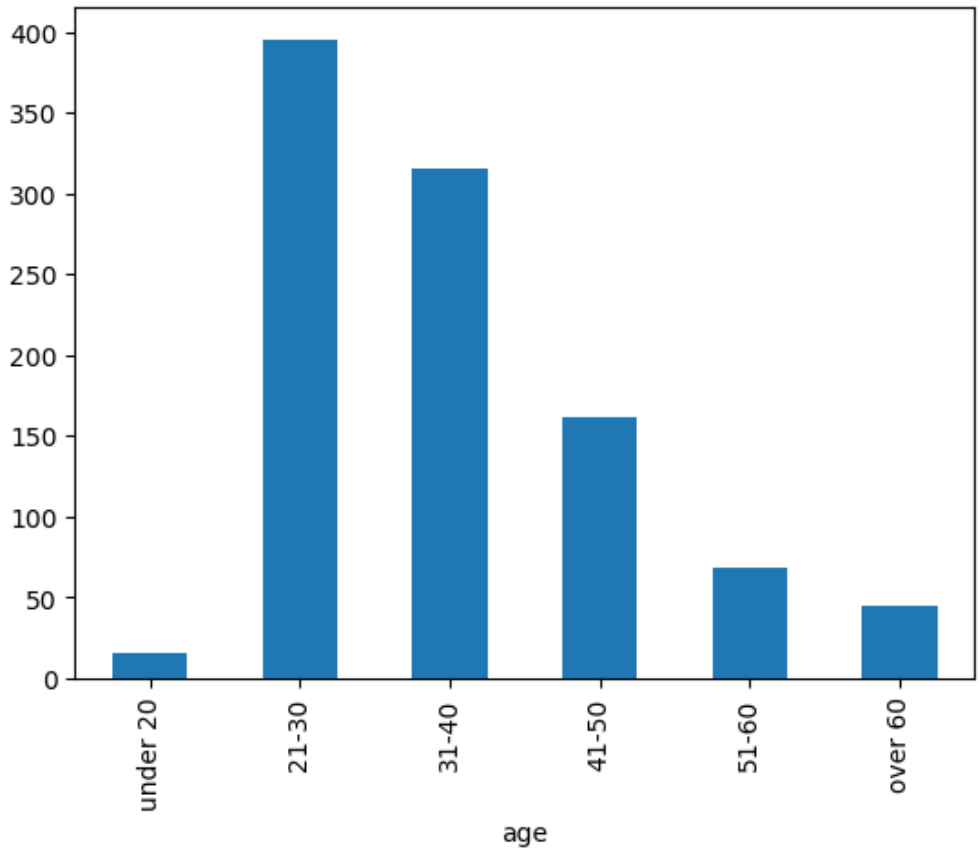
	count	mean	std	min	25%	50%	75%	max
Unnamed: 0	1000.0	500.500	288.819436	1.0	250.75	500.5	750.25	1000.0
Default	1000.0	0.300	0.458487	0.0	0.00	0.0	1.00	1.0
duration	1000.0	20.903	12.058814	4.0	12.00	18.0	24.00	72.0
amount	1000.0	3271.258	2822.736876	250.0	1365.50	2319.5	3972.25	18424.0
installment	1000.0	2.973	1.118715	1.0	2.00	3.0	4.00	4.0
residence	1000.0	2.845	1.103718	1.0	2.00	3.0	4.00	4.0
age	1000.0	35.546	11.375469	19.0	27.00	33.0	42.00	75.0
cards	1000.0	1.407	0.577654	1.0	1.00	1.0	2.00	4.0
liable	1000.0	1.155	0.362086	1.0	1.00	1.0	1.00	2.0

4. Age binning bar chart

To reduce the amount of personal detail in the age attribute, I grouped ages into bins such as 21–30, 31–40, and 41–50. This replaces exact ages with broader categories.Generalising age still allows us to see the general age patterns in the dataset without revealing exact ages.

```
[10]: gc["age"].value_counts(sort=False).plot(kind="bar")
```

```
[10]: <Axes: xlabel='age'>
```

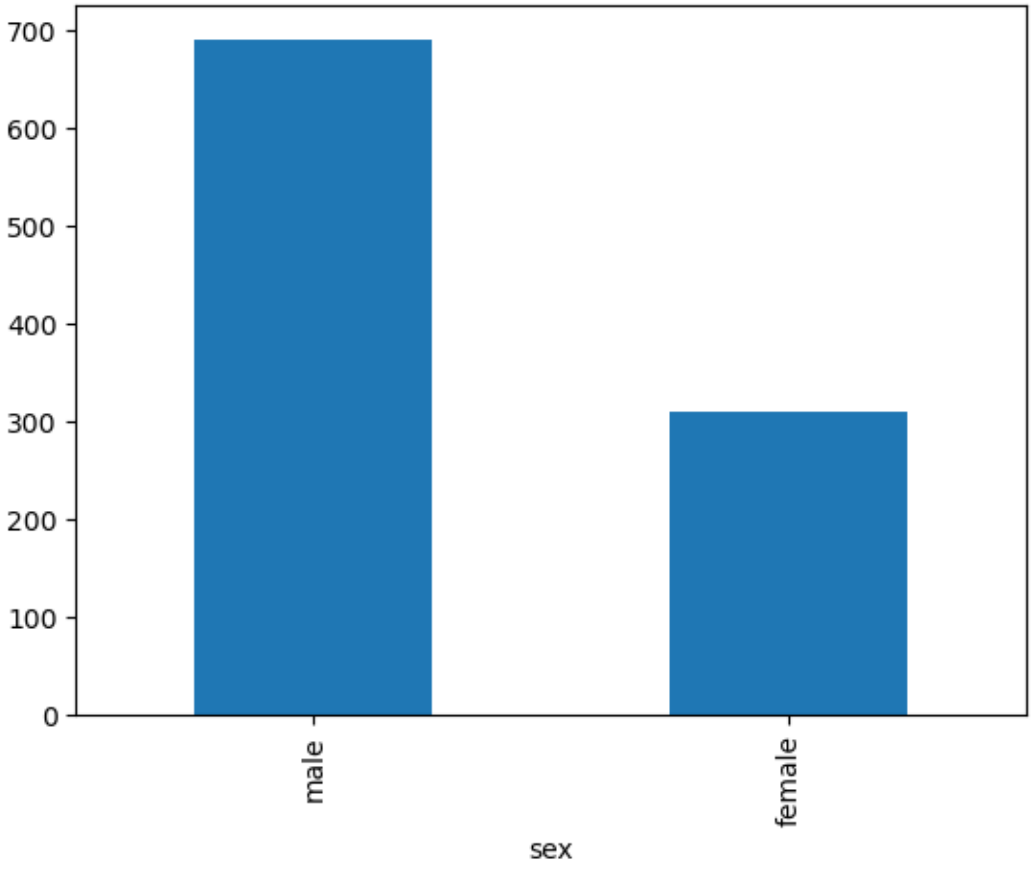


Sex bar chart before suppressed

Before removing the sex attribute, I plotted a simple bar chart to understand the distribution in the dataset.After that, I used suppression to remove the sex attribute completely.Suppression removes the whole attribute, which protects privacy but also limits how much we can analyse later.

```
[11]: gc["sex"].value_counts().plot(kind="bar")
```

```
[11]: <Axes: xlabel='sex'>
```



Dataframe after supressing sex

```
[12]: gc = gc.drop(columns=["sex"])
gc.head()
```

Reflection.

Working through this practical helped me understand how privacy techniques such as generalisation and suppression work in practice. I could clearly see how changing the detail in the data affects what can be learned from it.

Generalising the purpose and ageattributes reduced the amount of personal detail but still allowed useful patterns to remain. For example, it was still possible to see the most

common reasons for applying for a loan and which age groups appeared most often. This showed me that data can still be meaningful even when the exact values are removed.

Suppressing the sex attribute was more restrictive. Once it was removed, some comparisons were no longer possible. This helped me understand the trade-off between privacy and the usefulness of the dataset. Removing attributes increases protection but also limits what can be analysed. Seeing this directly in the notebook made the idea much clearer than just reading about it.

The practical also reminded me that datasets can still reveal information about groups of people even after anonymisation. This links back to the ethical topics in the lectures about responsibility, privacy, and handling human data carefully. Overall, working through the steps gave me a better understanding of why anonymisation matters and how it affects both privacy and analysis.